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Commensal Bacteria Modulate the Immune Response to Radiation Therapy (RT) in a Murine Oral Cavity Model

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Purpose/Objective(s): Oral cavity squamous cell carcinomas (OCSCC) carry high mortality and treatment morbidity. Unlike other mucosal subsites of the head and neck where definitive radiation can cure most cases, RT of gross disease in OCSCC is often ineffective. Thus, there is a need to better understand patient factors that influence response to RT in OCSCC. One such factor influencing response to treatment is the microbiome of the oral cavity and gut. Prior work has shown that the intestinal microbiome shapes systemic immunity and dictates responses to chemotherapy, immunotherapy and RT. Therefore, we hypothesized that the bacterial microbiome is an essential component mediating the response to RT in a murine model of OCSCC.

Materials/Methods: Mouse oral squamous cell carcinoma (MOC1 and MOC2) cells were injected subcutaneously into C57BL/6 mice. Once tumors were palpable, the mice were treated with a broad-spectrum antibiotic (Abx) cocktail of ampicillin, cilastatin, imipenem and vancomycin to deplete intestinal bacteria. At tumor volumes of $\sim$200 mm^3, tumors were focally irradiated (8 Gy single fraction) using the X-RAD SmART+ platform with image guidance. Tumor volumes were compared using repeated measures two-way ANOVA, and survival curves were analyzed using the log-rank test. We use 16S microbiome sequencing, high-dimensional flow cytometry and single-cell RNA sequencing to identify and phenotypically characterize immune cells in the tumor microenvironment for irradiated tumors in the presence and absence of the bacterial microbiome.

Results: In the immune-sensitive OCSCC model (MOC1), treatment with Abx prior to RT decreased the efficacy of RT as shown by faster tumor growth ($P<0.01$) and worse survival ($P<0.05$) in mice. In the immune-insensitive OCSCC model (MOC2), treatment with Abx prior to RT did not influence tumor growth or survival. MOC1 tumors exhibited more RT-elicited anti-tumor immunity following RT compared to MOC2, which was abrogated with the addition of Abx.

Conclusion: The bacterial microbiome regulates anti-tumor responses following RT in OCSCC tumors that are immune responsive, but not in those that are immune insensitive. Therefore, strategies targeting the microbiome rely on understanding the complex interplay between the microbiome, tumor and the immune system.

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Assessing Clinical and Dosimetric Benefits of Spot-Scanning Proton Arc Therapy in Ocular Melanoma Treatment

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Purpose/Objective(s): To evaluate the feasibility and dosimetric superiority of spot-scanning proton arc therapy (SPArc) for ocular melanoma, in comparison to Intensity Modulated Proton Therapy

Materials/Methods: Six ocular melanoma cases, varying in size (large/small) and anatomical locations (posterior/anterior/optic disc), were planned with 3-field IMPT (3F-IMPT) and SPArc for dosimetric comparison. With a target dose normalized to 50 cGE, 3.5% range uncertainty were used in both groups for robust optimization. Dose to organs-at risk (OARs) were analyzed. The mean dose averaged linear energy transfer (LET_d) were calculated with dose threshold of voxels receiving no less than 20% of the maximum dose.

Results: SPArc demonstrated a consistently superior dose distribution compared to 3F-IMPT in all cases. Notably, SPArc achieved improved dose sparing of lens and cornea in tumor situated in the posterior segment and optic disc region, where the lens and cornea were in the beam path. Specifically, for the four cases involving tumors in the posterior segment or optic disc, the median maximum dose (Dmax) to the lens with IMPT and SPArc was 30.61 cGy and 10.62 cGy, respectively. SPArc achieved a 65.3% reduction in the lens's Dmax. A similar outcome was observed for the mean dose (Dmean) to the cornea, with SPArc achieving a 76.7% reduction in the Cornea Dmean. Additionally, SPArc presented an increased LET_d within the target area while reducing the LET_d in OARs abutting the target. Particularly, the median of mean LETd for tumors was 3.90 keV/μm with IMPT compared to 4.14 keV/μm with SPArc. For the lens, the median of mean LETd was significantly lower with SPArc at 0.60 keV/μm, compared to 1.92 keV/μm with IMPT.

Conclusion: As an innovative proton therapy approach, SPArc without utilizing the compensator and blocks or MLCs has demonstrated dosimetric superiority and adaptability in treating ocular melanomas situated in the posterior segment and optic disc region, especially in scenarios where sparing of the lens and cornea is crucial.

Abstract 3617 − Table 1

Tumor Size/ Location	Plan	Cornea (mean) cGE	Lacrimal Gland (mean) cGE	Retina (mean) cGE	Optic Nerve 0.03cc (max) cGE	Lens 0.03cc (max) cGE
Anterior Large	3F IMPT SPArc	35.00 14.82	20.66 10.42	36.40 29.55	38.04 28.91	42.48 28.39
Anterior Small	3F IMPT SPArc	46.20 44.65	17.03 0.82	6.89 5.15	0.20 1.83	53.81 51.79
Optic Disc Large	3F IMPT SPArc	26.67 6.77	28.75 8.80	37.65 29.98	50.68 44.69	34.97 13.54
Optic Disc Small	3F IMPT SPArc	19.56 3.72	14.52 4.32	34.54 23.67	52.09 50.34	26.25 7.69
Posterior Large	3F IMPT SPArc	33.68 5.95	33.00 15.12	41.72 29.95	34.51 30.45	42.74 16.73
Posterior Small	3F IMPT SPArc	16.86 1.96	20.80 11.40	29.10 15.44	38.22 29.33	23.41 3.87

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